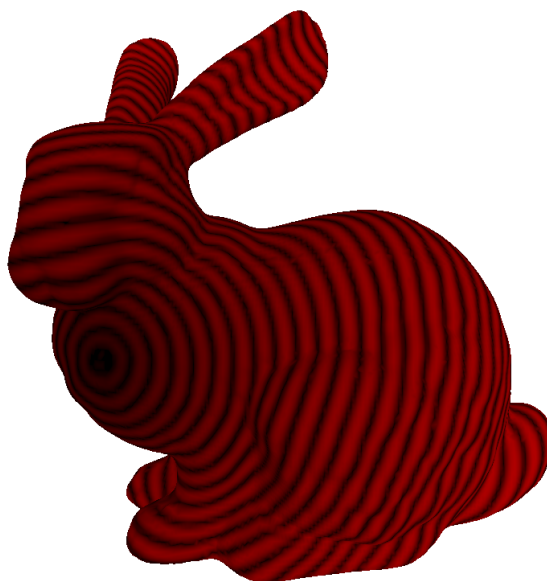


Sheet G12 - Heat Methods

Solutions for the theoretical and practical part via eCampus by Mo, 26.01.2026, **10:00**.

Practical Part



Assignment 1) Geodesics in Heat

(10Pts)

Download `framework_g12.zip` from eCampus and add it to the project. Implement the geodesics in heat algorithm [1]. As an additional resource: <https://www.youtube.com/watch?v=Dgu-V9ciGi8>.

- a) In `compute_matrices(...)` compute and factorize the operators $(\mathbf{A} - t\mathbf{L}_C)$ and $\mathbf{L}_C$.
- b) Solve a single step of the inverse Euler iteration using the precomputed matrices.
- c) Compute the gradient $\nabla \mathbf{u}$ for each triangle.
- d) Compute the divergence $\nabla \cdot \mathbf{X}$.
- e) Solve the Poisson equation $\Delta \phi = \nabla \cdot \mathbf{X}$.

Theoretical Part

Upload the solution as pdf file on eCampus!

Assignment 2) Green's function

(4Pts)

A Green's function is the solution to

$$\Delta G(x, y) = \delta(x - y)$$

Let $\{\phi_i, \lambda_i\}_{i \geq 1}$ be the eigendecomposition of Δ (i.e., $\Delta \phi_i(x) = \lambda_i \phi_i(x)$). Show that

$$G(x, y) = \sum_{i=1}^{\infty} \frac{\phi_i^*(x) \phi_i(y)}{\lambda_i}.$$

Write $G(x, y)$ as a linear combination of the eigenfunctions and apply the Laplace Operator to this expression. Use the fact that the eigendecomposition is complete (i.e., the basis functions span the whole function space):

$$\delta(x - y) = \sum_{i=1}^{\infty} \phi_i^*(x) \phi_i(y).$$

References

- [1] Keenan Crane, Clarisse Weischedel, and Max Wardetzky. "Geodesics in heat: A new approach to computing distance based on heat flow". In: *ACM Transactions on Graphics (TOG)* 32.5 (2013), pp. 1–11.

Good luck!